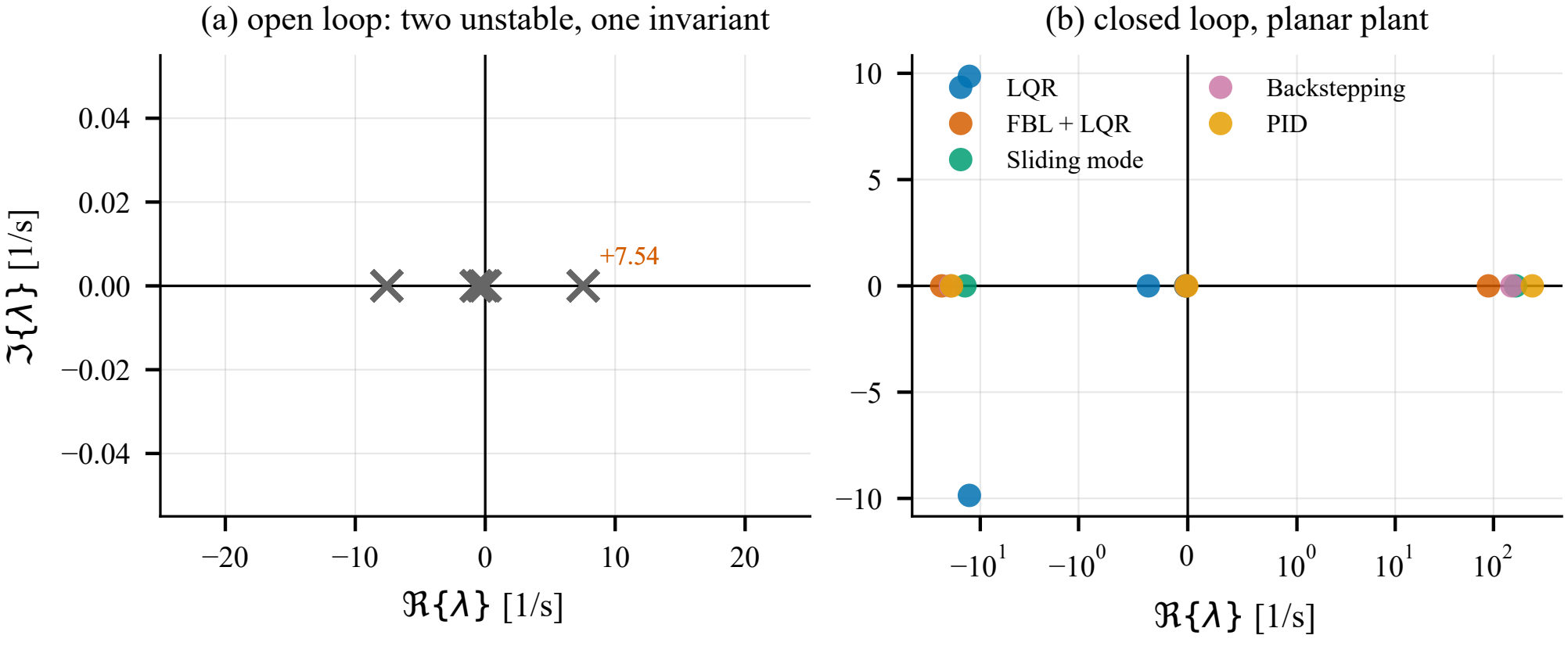
(a) open loop: two unstable, one invariant
0.04
0.02
0.00
−0.02
−0.04
+7.54
−20
−10
0
10
20
ℑ{λ} [1/s]
ℜ{λ} [1/s]
(b) closed loop, planar plant
LQR
FBL + LQR
Sliding mode
Backstepping
PID
10
5
0
−5
−10
−10^1
−10^0
0
10^0
10^1
10^2
ℜ{λ} [1/s]